C11PA Project Management 2026 Coursework 2

Group 21 Project

Project Theme: Biodegradable Packaging for a Retail Supermarket

Elected supermarket: TESCO – Corstorphine Extra



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1. Executive Summary

This project proposes the implementation of seaweed-based compostable packaging for 20 high-rotation ready-to-eat SKUs in Tesco Corstorphine Extra. It addresses the environmental impact of plastic packaging while aligning with Tesco's Planet Plan.

The pilot will run over 8 months with a £1.5M budget, focusing on operational deployment rather than material development. Key phases include supplier selection, packaging validation, certification, store rollout, and performance evaluation.

The project targets a minimum 60% reduction in plastic use while maintaining product availability above 90%, customer satisfaction above 90%, and packaging performance above 95%.

A clear governance and monitoring setup ensures control over delivery and supports future scaling across Tesco.

2. Project Scope

2.1 Project Purpose and Intended Outcome

The project focuses on the implementation of certified seaweed-based compostable packaging (EN 13432 compliant) for approximately 20 high-rotation Own Brand ready-to-eat SKUs within Tesco Corstorphine Extra, Edinburgh (full list provided in Appendix 2). The selection of 20 SKUs balances operational manageability with sufficient scale to generate statistically meaningful performance data.

The selected products prioritise high plastic usage, strong customer visibility, and operational feasibility within store-level replenishment processes. The project responds to the continued reliance on fossil-based plastic packaging in high-volume ready-to-eat categories, which contributes to resource depletion, carbon emissions, and waste management challenges (OECD, 2022) (UNEP, 2023)

Although current UK retail packaging is largely recyclable, its effectiveness depends on consumer behaviour and infrastructure consistency (WRAP, 2021). This project therefore explores compostable alternatives as a complementary solution within a circular economy framework.

The initiative aligns with Tesco's Planet Plan commitments, including the target of fully recyclable Own Brand packaging by 2025 and broader sustainability objectives (Tesco, 2025) piloting compostable packaging within a defined category, the project supports Tesco's packaging transformation strategy.

The intended outcome is the successful operational integration of compostable packaging for selected SKUs, replacing conventional plastic materials while maintaining food safety, shelf-life performance, and operational efficiency. The project is structured as an implementation initiative rather than material development, ensuring feasibility within existing retail operations.

At completion, the project will deliver operational compostable packaging, verified EN 13432 compliance, measurable plastic reduction, and a monitoring framework to support scaling decisions.

2.2 Project Output and Key Characteristics

The primary output is the full deployment of seaweed-based compostable packaging for selected ready-to-eat products in the pilot store.

The packaging must meet the following characteristics:

- Certified industrial compostability under EN 13432 (CEN, 2000)
- Compliance with UK food-contact regulations (UK Government, 2019)
- Adherence to Food Standards Agency guidance (Agency, 2023)
- Functional equivalence to plastic packaging in durability and protection
- Preservation of shelf-life standards

- Clear sustainability labelling
- Compatibility with Tesco's store operations

The project excludes redesign of food products and focuses solely on packaging transition.

2.3 Identification of End Users

The primary end users are Tesco customers purchasing ready-to-eat products in the pilot store.

Operational users include store staff responsible for handling and replenishment, whose processes may be affected by changes in packaging format.

Strategic users include Tesco procurement and sustainability teams, who rely on performance data, supplier compliance, and cost implications to inform scaling decisions.

End-of-life stakeholders include waste management actors responsible for composting processes, which are critical to achieving the intended environmental benefits.

The solution must therefore satisfy customer usability, operational feasibility, and end-of-life system compatibility.

2.4 Scope Boundaries

IN SCOPE	OUT OF SCOPE
Implementation of compostable packaging for approximately 20 selected SKUs	Development of new biopolymer materials
Supplier contracting and integration	Construction of manufacturing facilities
Technical validation and performance testing	Nationwide rollout beyond the pilot store
Regulatory compliance verification	Modification of food product recipes
Store-level deployment	Structural changes to Tesco's national logistics system
Staff training	
Pilot monitoring and evaluation	

Table 1: Project Scope Boundaries [Source: Own Elaboration]

These scope boundaries clearly define project inclusions and exclusions, ensuring controlled delivery and preventing scope creep.

2.5 Link to the Sustainability Challenge

Plastic packaging remains a central issue within Tesco's sustainability strategy. While 87% of Own Brand packaging is recyclable at kerbside, rising to 99% including collection points, reliance on fossil-based materials persists in high-volume categories (Tesco, 2025)

By implementing compostable packaging within a defined product category, this project contributes to reducing single-use plastics, advancing bio-based materials, and improving end-of-life outcomes. It aligns with SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action) (United Nations, 2015)

The defined scope ensures that sustainability objectives are translated into a measurable and governed project outcome.

3. Project Objectives

The objective of this project is to design and implement a controlled pilot deployment of certified seaweed-based compostable packaging for 20 selected ready-to-eat SKUs within Tesco Extra Corstorphine, ensuring measurable environmental impact while maintaining commercial and operational performance.

Business Objective

To successfully implement the packaging transition across 20 high-rotation SKUs within an 8-month project lifecycle, achieving zero disruption to product availability and maintaining customer satisfaction levels above 90% during the pilot period.

Environmental Objective

To reduce conventional single-use plastic packaging across the selected SKUs by at least 60% by weight during the pilot phase, contributing to Tesco's Planet Plan and circular economy commitments. The packaging must be certified industrially compostable under EN 13432 and compatible with existing UK waste processing infrastructure.

Technical Objective

To ensure the seaweed-based packaging achieves full functional equivalence with existing plastic packaging, including durability, food safety compliance, moisture resistance and preservation of shelf-life standards. Certification under EN 13432 and compliance with UK food-contact regulations must be obtained prior to store rollout.

Cost Objective

To implement the pilot within the allocated sustainability innovation budget, ensuring that incremental packaging cost per SKU does not exceed a 10% variance compared to current packaging costs, while identifying potential long-term scale efficiencies.

Schedule Objective

To complete supplier selection, packaging validation, certification, and store rollout within an 8-month timeframe, with clearly defined milestone gates for design approval, compliance validation and operational launch.

Quality Objective

To achieve a minimum 95% packaging performance success rate during the first three months of rollout, measured through defect rates, handling performance and customer feedback.

Intended Benefits and Outcomes

For Tesco (Project Sponsor), the pilot provides measurable progress toward sustainability targets, reputational enhancement, and a scalable model for wider packaging transformation.

For operational stakeholders, it ensures structured implementation with minimal disruption and clear governance.

For customers and environmental stakeholders, it delivers a visible reduction in plastic usage and improved sustainability transparency.

Cost control is prioritised to ensure scalability, even if it limits the speed of material optimisation.

4. Project Deliverables and Schedule

4.1 Project Deliverables

The project will produce 5 major deliverables across its 8-month lifecycle. Each deliverable corresponds to a defined phase of work, has measurable outputs and is subject to formal gate review prior to progression.

Code	Deliverable	Key Output	Gate	Gate Date
D1	**Start of project and supplier selection**	Signed supplier contract	Gate 1	30 Jun 2026
D2	**Packaging Design & Approval**	Approved prototype designs, functional testing report	Gate 2	12 Aug 2026
D3	**Certification of packaging to ensure compliance with EN 13432**	EN 13432 certificate, UK food-contact compliance	Gate 3	18 Sep 2026
D4	**Store Rollout & Staff Training**	Live deployment across 20 SKUs, trained store staff	Gate 4	28 Nov 2026
D5	**Evaluation and Feedback**	KPI performance report, scaling recommendation	Gate 5	30 Jan 2027

Table 2 Project Deliverables Overview [Source: Own Elaboration]

D1 – Project Initiation and Supplier selection (June- July)

This deliverable secures a certified packaging supplier through supplier market analysis and shortlisting and finally contract negotiation and signing with a supplier capable of delivering seaweed-based packaging certified to EN 13432. This Deliverable is foundational to all subsequent phases as supplier capability directly determines technical feasibility, regulatory compliance timelines and cost performance.

D2 – Packaging Design and Approval (July - August)

This deliverable covers the development and approval of the compostable packaging solution for each of the 20 selected SKUs. Supplier will produce prototypes against agreed specification including durability, food safety compliance, shelf-life preservation and sustainability labelling. An internal review and functional testing process will be conducted prior to the formal design approval gate.

D3 – Certification & Regulatory Compliance (August – September)

This deliverable ensures that the packaging meet regulatory and EN 13432 compostability standards before in-store deployment. No packaging will progress without approval from Gate 3.

D4 - Store Rollout and Staff Training (September - November)

The largest deliverable encompasses the full deployment of the compostable packaging across all 20 target SKUs. Logistics and supply chain integration, staff training programme, stacking

stability, storage conditions and end-of-life waste management and phased instore deployment across all 20 SKUs. The gate confirms that all SKUs are live and product availability has been maintained above 90% threshold defined in the project objectives.

D5 - Project Monitoring and Evaluation (November - January)

The filth deliverable captures pilot project performance and produces evidence for scaling decision. KPI and performance tracking, customer feedback analysis run parallel and a final report with a scaling recommendation submitted to the sponsor.

4.2 Work Breakdown Structure

The WBS is structured across five work packages corresponding directly to the five deliverables, with each package decomposed into discrete activities assigned to responsible parties with timing. The full break down is provided in Appendix A4.2.

4.3 Cost Breakdown Structure

The CBS allocates the total project budget of £1.5 million (£ 1,500,000) across five work packages, establishing a transparent cost monitoring framework, variance analysis and informed decision making throughout the project lifecycle. A contingency fund of 10% of the budget (£150,000) is to be maintained and will be accessible upon approval from the project sponsor to ensure disciplined budget control. The full project cost plan is provided in Appendix A4.3.

4.4 Project Schedules and Milestones

The project is scheduled across eight months from June 2026 to January 2027. Five formal milestone gates are embedded, each requiring approval before the subsequent phase is initiated. Gate 1 (30 June 2026) confirms that a qualified supplier has been contracted and the project is fully operational. Gate 2 (12 August 2026) validates the packaging prototypes meet all functional requirements before committing to certification investment. Gate 3 (18 September 2026) provides the critical regulatory assurance required before any packaging enters the store floor. Gate 4 (28 November 2026) confirms the full in-store deployment and the operational readiness across all 20 SKUs. Gate 5 (30 January 2026) marks the formal conclusion of the project. The project will be evaluated and a report and the viability and the scalability of the project will be submitted to the project sponsor. The schedule incorporates buffer periods between phases to accommodate potential delays in supplier response times, certification processing and logistics coordination. The Contingency reserve will be used to address cost impacts arising from schedule slippage. The full Gantt chart developed in Microsoft Project with all WBS codes, resource assignments and critical path is provided in Appendix A4.4.



Figure 1 WBS and CBS [Source: Own Elaboration]

5. Project Constraints and Assumptions

5.1 Project Constraints

The project is constrained by time, cost, technical, regulatory, and operational factors. It must be delivered within an 8-month timeframe (June 2026 – January 2027), with certification acting as a critical milestone that directly impacts subsequent phases such as store rollout.

Financially, the budget is capped at £1.5 million, including contingency, limiting flexibility in supplier selection and material iterations. From a technical perspective, the packaging must match the performance of plastic solutions in durability, moisture resistance, and shelf-life, as any failure would affect both quality KPIs and customer acceptance.

Additionally, all solutions must comply with EN 13432 and UK food-contact regulations, with certification timelines controlled externally. Finally, the packaging must integrate seamlessly into Tesco's existing store operations without requiring structural changes to logistics or supply chain systems.

5.2 Project Assumptions

The project assumes the availability of a supplier capable of delivering certified seaweed-based packaging at the required scale and within the defined timeline. It is also assumed that the material will meet technical performance requirements, including durability and food safety, following testing and validation.

From a market perspective, it is expected that customers will accept the new packaging without negatively impacting purchasing behaviour, provided product quality remains unchanged. Additionally, the UK waste management infrastructure is assumed to be sufficiently developed to process compostable materials, allowing the intended environmental benefits to be realised.

The project further assumes a stable regulatory environment, with no major changes affecting compostability or food-contact standards during the lifecycle. Finally, it relies on alignment and cooperation across Tesco's internal teams to support implementation within the planned timeframe.

The project assumes stable supplier performance, although variability in bio-based materials remains a potential limitation.

6. Stakeholder Analysis and Management

6.1 Stakeholder Identification

The implementation of certified seaweed-based compostable packaging in Tesco Corstorphine Extra involves several internal and external stakeholders who influence, or are influenced by, the project. As the initiative requires coordination across multiple departments and external partners, identifying key stakeholders is essential to ensure effective governance, minimise risks, and support successful implementation.

Stakeholders were identified based on their level of influence over project decisions, their exposure to project outcomes, and their operational or strategic involvement.

INTERNAL STAKEHOLDERS

1.Tesco Sustainability Director (Project Sponsor)

Acts as the executive sponsor and ensures alignment with Tesco's Planet Plan governance framework and circular economy targets (Tesco, 2025). The sponsor holds high decision-making authority and oversees sustainability performance.

2.Project Manager

Responsible for overall project coordination, planning, stakeholder integration and risk management, reporting directly to the Project Sponsor.

3.Tesco Procurement Department

Manages supplier selection, contract negotiation and cost control, playing a key role in securing the packaging supplier.

4.Category Manager – Ready-to-Eat Products

Oversees category performance and supplier relationships, ensuring that product quality and customer satisfaction are maintained during the transition.

5.Store Manager and Store Operations Staff

Responsible for operational implementation within the store, including product handling, stocking and monitoring of the new packaging system.

6.Finance Department

Monitors budget compliance and financial performance of the pilot project.

EXTERNAL STAKEHOLDERS

1.Seaweed-Based Packaging Supplier

Provides the certified compostable packaging and ensures material performance, regulatory compliance and reliable delivery.

2.Waste Management Partner

Supports the waste processing infrastructure aligned with Tesco's waste reduction commitments

3.Customers

Their acceptance and purchasing behaviour will influence the commercial success of the pilot.

4.Regulatory Authorities

Ensure compliance with UK food-contact safety and environmental regulations.

6.2 Stakeholder Power-Interest Analysis

To prioritise stakeholder engagement and allocate management resources effectively, stakeholders have been analysed using the Power–Interest Matrix, a tool that classifies stakeholders according to their level of influence over project decisions and their interest in project outcomes.

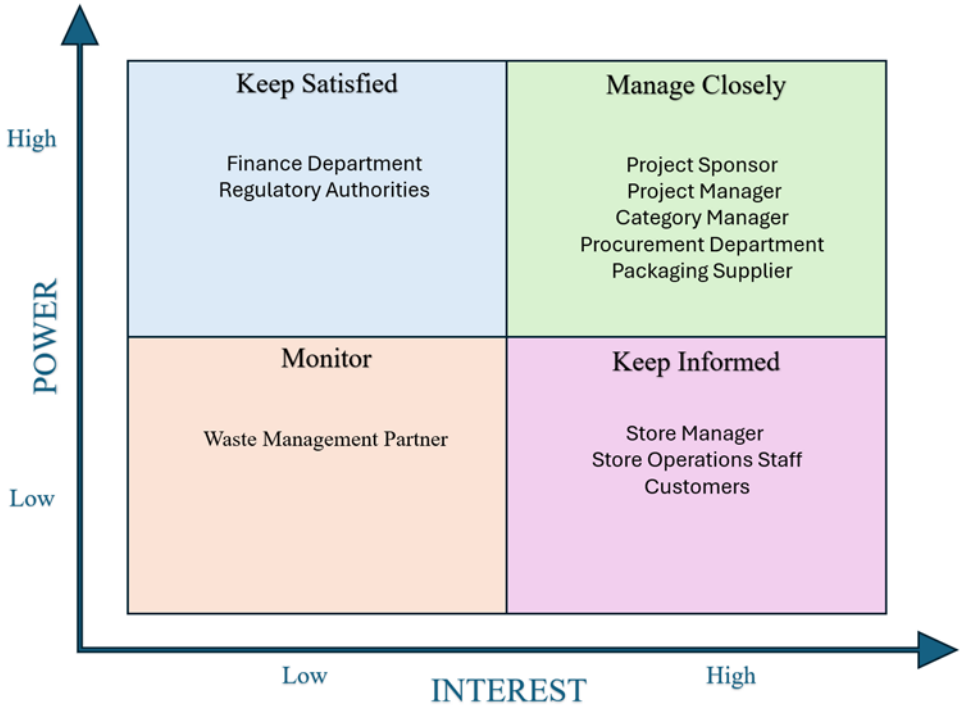


Figure 2 Stakeholder Power-Interest Matrix [Source: Own Elaboration]

Stakeholders with high power and high interest, such as the Project Sponsor, Project Manager and Procurement Department, require close management because they play a critical role in decision-making, supplier contracting and project governance.

Stakeholders with high power but lower operational involvement, including the Finance Department and Regulatory Authorities, must be kept satisfied through regular reporting and compliance assurance.

Operational stakeholders such as the Store Manager, store staff and customers have high interest but lower formal power, therefore they should be kept informed and actively engaged during implementation.

6.3 Project responsibility Matrix (RACI)

Effective stakeholder engagement is critical for the successful implementation of the biodegradable packaging pilot, as the project requires coordination between multiple internal departments and external partners. Engagement strategies are therefore designed according to stakeholders' level of power and interest, ensuring that communication and involvement are proportionate to their influence on project outcomes.

High-power stakeholders, including the Project Sponsor, Procurement Department and Category Manager, will be engaged through regular governance meetings and progress reports to support strategic oversight and decision-making. Operational stakeholders such as the Store Manager and store staff will receive implementation briefings and training to ensure smooth deployment of the new packaging system.

These engagement activities are supported by the project governance framework. The **Responsibility Assignment Matrix (RACI)** defines clear roles and accountability across project activities, while the **Communication Plan** establishes structured information flows between stakeholders. In addition, the **Organisational Breakdown Structure (OBS)** clarifies reporting relationships between internal teams and external partners. Together, these tools support transparency, coordination and effective decision-making throughout the project lifecycle. These three frameworks are presented in the Appendixes.

7. Critical Success Factors (CSFs) and Key Performance Indicators (KPIs)

The success of the biodegradable packaging pilot at Tesco Corstorphine Extra depends on clearly defined Critical Success Factors (CSFs) supported by measurable KPIs, creating a structured framework for performance monitoring, compliance, and data-driven decision-making.

Environmental performance is a key CSF, focused on reducing single-use plastic in ready-to-eat products. KPIs include the percentage reduction in plastic packaging and the volume of compostable materials introduced, demonstrating sustainability impact.

Technical and regulatory compliance ensures packaging meets EN 13432 standards and UK food-contact regulations. KPIs track certification rates, compliance incidents, and performance factors such as durability and shelf life to maintain quality standards.

Operational efficiency is monitored through KPIs such as milestone completion, staff training rates, and operational error levels, assessing how effectively the new packaging integrates into store processes.

Financial performance requires the project to remain within the $1.5 million budget while maintaining cost efficiency. KPIs include cost variance, unit procurement costs, and total expenditure.

Stakeholder acceptance is also critical, measured through customer satisfaction, product sales, and staff compliance, reflecting both market response and internal adoption.

Together, these CSFs and KPIs enable continuous monitoring, early issue detection, and informed decision-making to support potential scalability across Tesco stores.

8. Risk Management

8.1 Framework and Strategic Approach

Our project adopts a proactive risk management strategy based on the APM Body of Knowledge (7th edition). Instead of simply reacting to issues, we identify and manage threats early to protect the project's "Triple Constraint":

- Time: Completing the pilot within the 8-month schedule (June 2026 – January 2027).
- Cost: Managing the project within the total budget of £1,500,000.
- Quality: Meeting high safety and performance standards for food packaging.

By using our Responsibility Matrix (Section 6.3), we have assigned a "Risk Owner" to every identified threat. This ensures clear accountability between the Tesco project team and external suppliers, allowing for rapid response to deviations (Association for Project Management, 2019)

8.2 Analysis of Key Risk Categories

To ensure the project meets its objectives, we have analysed the following critical areas:

8.2.1 Government & Food Regulations

- **Food Safety:** According to the Materials and Articles in Contact with Food Regulations 2012, we must guarantee zero chemical migration from the seaweed film into food products (Food Standards Agency, 2020). We will run rigorous tests during Phase 3 (Testing & Validation) to ensure the seaweed film is 100% safe for Tesco's products before the Gate 3 approval on September 25th.

- **Chemical Restrictions:** We are actively monitoring two key safety lists from the Health and Safety Executive (HSE, 2026)

    - The Authorisation List (Annex XIV): Chemicals that are banned unless a special permit is granted.
    - The List of Restrictions: Chemicals that have very strict limits on how they can be used.

- **Impact on Project:** If any substance used to stabilize our seaweed is added to these lists as a Substance of Very High Concern (SVHC), Tesco Legal and Procurement will immediately trigger a change in our material provider. This proactive monitoring ensures the project stays legal and safe for Tesco customers throughout the delivery timeline ending in January 2027.

8.2.2 Packaging Durability

- **Moisture Sensitivity:** Seaweed-based materials are naturally vulnerable to water and humidity. Controlling this is essential to preserve packaging strength (Nesic, 2024)
- **KPI Impact:** Because Tesco's ready meals contain high moisture, material failure could cause leaks and food waste, risking our 95% Quality KPI and the "Little Helps Plan" (Tesco PLC, 2017). To prevent this, the Seaweed-based Packaging Supplier will run rigorous stress tests in Phase 2 to ensure structural integrity before mass production.

8.2.3 Schedule and Budget Safety

The certification process in Deliverable 3 (Testing & Validation) is the main schedule risk. Any delay in obtaining the EN 13432 compostability standard beyond September 25th (Gate 3) would act as a bottleneck, directly delaying the In-store deployment (Deliverable 4) and threatening our final completion date of January 15th, 2027.

To protect the project's financial health, we have included a £150,000 contingency fund (10% of the budget) on top of our £1,350,000 base cost. This buffer, managed by the Project Manager in consultation with the Sponsor, is reserved for unexpected issues like laboratory re-testing or raw material price spikes, ensuring the total investment remains within the £1.5M limit.

8.3 Risk Register Summary

To determine the priority of each risk, we have applied a Quantitative Risk Analysis based on a 5x5 matrix (see Appendix 8.1 for full definitions). Each event is scored by multiplying its Probability (1-5) and Impact (1-5). This scoring system ensures that high-impact threats to the £1.5M budget and the timeline ending January 15th, 2027, are identified early and assigned a specific Contingency Plan.

To ensure the project remains on track, we have prioritised the most critical threats based on their impact on our 5 project KPIs. Risks with a score of 12 or higher (R2 and R3) are classified as "Critical" and require weekly monitoring by the Project Manager. Furthermore, our financial planning incorporates a 10% contingency (£150,000) specifically designed to absorb technical (R2) and inflationary (R5) risks without exceeding the total investment limit.

The following table summarises the top risks and our strategic response:

ID	Project Risk	Score	Risk Owner	Mitigation / Response Plan
R1	**Legal & Food Safety**	9	Tesco Legal	Continuous monitoring of HSE (2026) lists and FSA 2012 compliance to ensure zero chemical migration.
R2	**Material Failure**	12	Seaweed-based Packaging Supplier	Intensive stress tests during Phase 2. If 95% quality is not met, switch to the pre-approved secondary provider.
R3	**Certification Delay**	12	Project Manager	Close follow-up with EN 13432 authority to meet Gate 3 (Sept 25th) and avoid delaying store rollout.
R4	**Operational Issues**	6	Tesco Corstorphine Store Manager	Staff training conducted in Phase 4 to ensure correct handling and replenishment of the 20 new SKUs.
R5	**Supply Chain Costs**	8	Tesco Procurement	Use of fixed-price contracts and the £150,000 contingency fund to absorb raw material price spikes.

Table 3 Summary of Critical Project Risks and Mitigation Strategies. [Source:Own elaboration]

Technical Note: All strategic responses for R2 (Material Failure) are supported by the research of Nesic et al. (2024) regarding the moisture-resistant properties of seaweed biopolymers. For the full technical breakdown, scoring scales, and risk matrix visualization, please refer to Appendix 8.

8.4 Risk Monitoring and Control

To keep the project on track, our Risk Register (Appendix 8) is not just a static list; it's a "live" management tool used throughout the project's lifecycle.

Every week, the Project Manager will check the "Triggers" (early warning signs) for each risk to spot potential issues before they become real problems. If a risk hits the "Critical Zone" (a score of 12 or more, such as R2-Material or R3-Certification), we will immediately escalate it to the Steering Committee as per our RACI matrix.

This process allows us to make fast, data-backed decisions on when to use the £150,000 contingency fund. By staying proactive, we ensure that the £1.5M investment is protected and that Tesco's seaweed-based packaging is ready for January 15th, 2027, completion (Gate 5).

9. Monitoring and Control

9.1 Monitoring Strategy and Tools

To ensure professional oversight, a Central Decision Dashboard has been developed (see Appendix 9), including a simulated status update for November 2026. This supports analysis of budget variances and KPI performance ahead of the final handover in January 2027.

To meet the January 15th deadline within the £1.5M budget, the project follows Harold Kerzner's principle that cost and schedule management must be integrated. Monitoring both simultaneously ensures the seaweed packaging rollout stays on track. (Kerzner, 2022)

- **MS Project Tracking:** Microsoft Project is used to manage tasks, with weekly updates from the Project Manager to track progress and identify delays, particularly in Phase 3 (Certification).
- **Visual Performance:** A Project Dashboard (Appendix 9) uses a traffic-light system to monitor five key KPIs, enabling quick assessment of budget and quality without excessive technical detail.
- **Quality Gates:** Five formal approval gates act as control checkpoints. Each phase requires Sponsor approval before proceeding, ensuring full certification prior to rollout at Tesco Corstorphine.

9.2 Communication Framework

To ensure efficiency, the communication plan follows Harold Kerzner's principles of minimising unnecessary meetings and providing the Sponsor with strategic, not technical, information. Meetings are limited to those essential for maintaining progress toward the January 15th deadline.

The reporting structure includes:

- **Weekly Progress Meetings:** Short updates between the Project Manager, Supplier, and Tech Team to review tasks and address any immediate risks.
- **Monthly Board Reviews:** Formal sessions presenting the Project Dashboard, summarising budget status and key KPIs.
- **Gate Approvals:** End-of-phase checkpoints where the Sponsor reviews progress and authorises the next stage, with decisions recorded in a formal log.

This approach ensures clear communication, reduces overload, and keeps key departments involved only in critical decisions, maintaining project focus and agility

9.3 Project Evaluation and Success Criteria (CSFs)

To determine the success of the seaweed packaging pilot, results will be measured against the Critical Success Factors (CSFs) tracked in the Dashboard. Evaluation at the end of Phase 5 includes:

- **Technical & Quality Success:** Achievement of the 95% moisture resistance target, with current results at 96.2%, confirming technical performance.
- **Regulatory & Environmental Goal:** Successful attainment of EN 13432 certification during testing, validating sustainability.
- **Financial Performance:** Assessment of staying within the £1.5M budget, including effective use of the £150,000 contingency (with £70,000 already used).
- **Customer & Business Impact:** Final feedback to confirm suitability for a UK-wide rollout.

These criteria ensure a balanced evaluation of technical, financial, regulatory, and market outcomes

10. Project Closure

The closure of the seaweed-based packaging pilot at Tesco Extra Corstorphine goes beyond ending implementation, ensuring proper evaluation, responsibility transfer, and long-term value creation.

The process begins with formal sponsor approval, confirming that all deliverables have been met: rollout across 20 SKUs, compliance with EN 13432 standards, and achievement of key performance indicators. Any deviations are documented for future improvement.

A structured evaluation compares planned and actual performance in scope, time, cost, and quality, while verifying product integrity, food safety, and operational integration. Financial performance is also reviewed to confirm economic viability.

Stakeholder feedback is gathered from internal teams and customers to assess both operational feasibility and market acceptance. Contractual and administrative aspects are formally closed, ensuring all obligations, payments, and documentation are completed and archived.

A formal handover transfers responsibility to operational teams, supported by documentation and training. Lessons learned are captured to support continuous improvement, and a final report summarises outcomes, financial results, and recommendations, with success ultimately measured by scalability and balance between sustainability and operational constraints.

Appendixes:

Note: The number and subindex accorded to each appendix is related to the section in the report with the same number and index

Appendix 2:

A.1 Selected Pilot SKUs

The following SKUs have been selected based on high plastic usage, customer visibility, and operational feasibility within the pilot scope.

NO.	SKU NAME	CATEGORY
1	Tesco Chicken Caesar Salad	Salad
2	Tesco Falafel & Houmous Salad	Salad
3	Tesco Egg & Cress Sandwich	Sandwich
4	Tesco Chicken Club Sandwich	Sandwich
5	Tesco Tuna Mayo Sandwich	Sandwich
6	Tesco Pasta Salad (Tomato & Mozzarella)	Pasta
7	Tesco Sushi Selection	Sushi
8	Tesco Wrap (Chicken & Bacon)	Wrap
9	Tesco High Protein Chicken Salad	Salad
10	Tesco High Protein Snack Pot	Snack
11	Tesco Gut Sense Salad Bowl	Salad
12	Tesco Fruit Pot	Fruit
13	Tesco Mixed Berry Pot	Fruit
14	Tesco Finest Chicken Katsu	Ready Meal
15	Tesco Finest Pasta Ready Meal	Ready Meal
16	Tesco Finest Italian Salad Bowl	Salad
17	Tesco Finest Couscous Bowl	Grain Bowl
18	Tesco Snack Veg Pot	Snack
19	Tesco Protein Yogurt Pot	Dairy
20	Tesco Fresh Prepared Fruit Mix	Fruit

Figure 3 Selected Pilot SKUs [Source: Own Elaboration]

Appendix 4:

A4.2 Work Breakdown Structure (WBS)

Deliverable	WBS	Activity / Work Package	Responsible	Timing
D1	**Start of project and supplier selection**			
	1.1	Project Start	PM / Sponsor	01/06/2026 – 05/06/2026
	1.2	Supplier Shortlisting	PM / Procurement	08/06/2026 – 19/06/2026
	1.3	Supplier Evaluation and Selection	PM / Procurement	22/06/2026 – 01/07/2026
	1.4	Contract Negotiation and signing	Procurement / Supplier	02/07/2026 – 10/07/2026
	1.5	***Gate -1 Supplier Approved***	PM / Sponsor	**10/07/2026**
D2	**Packaging Design and Approval**			
	2.1	Packaging design and specification	PM / Supplier	13/07/2026 – 17/07/2026
	2.2	Prototype design and sampling	Supplier	20/07/2026 – 07/08/2026
	2.3	Testing and Feedback	PM / Category Mgr	10/08/2026 – 21/08/2026
	2.4	***Gate 2 - Design approved***	PM / Sponsor	**21/08/2026**
D3	**Certification of packaging to ensure compliance with EN 13432**			
	3.1	EN 13432 biodegradability testing	Supplier / Cert. Body	24/08/2026 – 18/09/2026
	3.2	UK food contact compliance review	PM / Legal	24/08/2026 – 18/09/2026
	3.3	***Gate 3 - Certification Received***	PM / Sponsor	**18/09/2026**
D4	**Store Rollout and Staff Training**			
	4.1	Supply Chain integration	Procurement / Supplier	21/09/2026 – 02/10/2026
	4.2	Staff Training	Store Manager / PM	05/10/2026 – 16/10/2026
	4.3	In store deployment (20 SKU)	Store Manager / Ops	19/10/2026 – 27/11/2026
	4.4	***Gate 4 - Rollout complete***	PM / Sponsor	**27/11/2026**
D5	**Evaluation and Feedback**			
	5.1	KPI data collection & performance tracking	PM / Store Ops	30/11/2026 – 01/01/2027
	5.2	customer feedback and analysis	PM / Category Mgr	30/11/2026 – 01/01/2027
	5.3	Final evaluation report & recommendations	PM / Sponsor	04/01/2027 – 15/01/2027
	5.4	***Gate 5 - Project Complete***	Sponsor	**15/01/2027**

Table 4 WBS [Source: Own Elaboration]

A4.3 Cost Breakdown Structure (CBS)

Deliverable	WBS	Activity / Work Package	% of budget	Cost
D1	**Start of project and supplier selection**		**3.00%**	**£ 45,000.00**
	1.1	Project Start	0.33%	£ 5,000.00
	1.2	Supplier Shortlisting	0.67%	£ 10,000.00
	1.3	Supplier Evaluation and Selection	1.00%	£ 15,000.00
	1.4	Contract Negotiation and signing	1.00%	£ 15,000.00
	1.5	***Gate -1 Supplier Approved***		
D2	**Packaging Design and Approval**		**18.66%**	**£ 280,000.00**
	2.1	Packaging design and specification	1.33%	£ 20,000.00
	2.2	Prototype design and sampling	12.00%	£ 180,000.00
	2.3	Testing and Feedback	5.33%	£ 80,000.00
	2.4	***Gate 2 - Design approved***		
D3	**Certification of packaging to ensure compliance with EN 13432**		**8.00%**	**£ 120,000.00**
	3.1	EN 13432 biodegradability testing	5.33%	£ 80,000.00
	3.2	UK food contact compliance review	2.67%	£ 40,000.00
	3.3	***Gate 3 - Certification Received***		
D4	**Store Rollout and Staff Training**		**50.00%**	**£ 750,000.00**
	4.1	Supply Chain integration	13.33%	£ 200,000.00
	4.2	Staff Training	3.33%	£ 50,000.00
	4.3	In store deployment (20 SKU)	33.34%	£ 500,000.00
	4.4	***Gate 4 - Rollout complete***		
D5	**Evaluation and Feedback**		**10.34%**	**£ 155,000.00**
	5.1	KPI data collection & performance tracking	4.00%	£ 60,000.00
	5.2	customer feedback and analysis	2.67%	£ 40,000.00
	5.3	Final evaluation report & recommendations	3.67%	£ 55,000.00

	5.4	*Gate 5 - Project Complete*		
Contingency Fund			**10%**	**£ 150, 000.00**

Table 5 CBS [Source: Own Elaboration]

A4.4 Gantt Chart

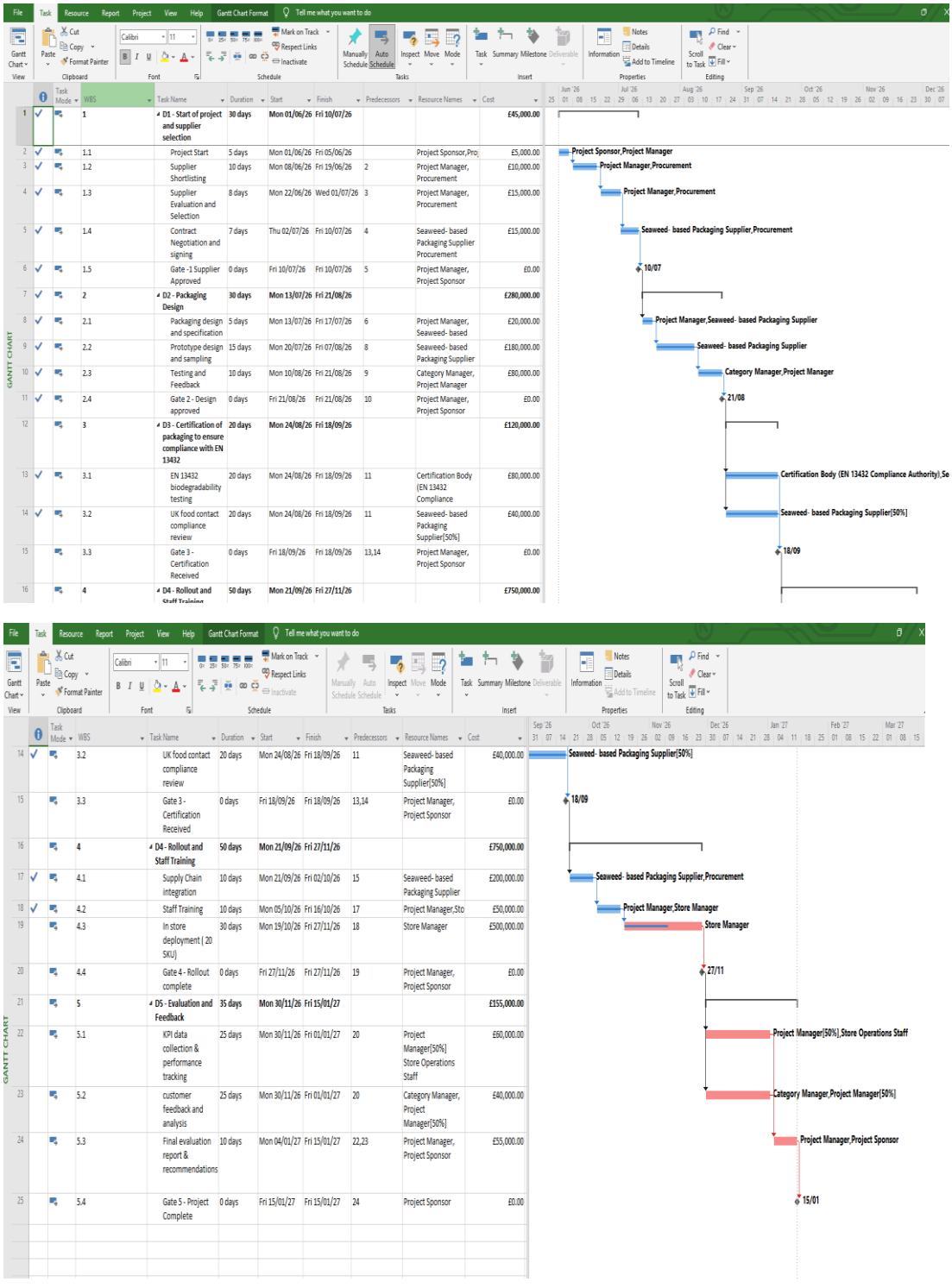


Figure 4 Gantt Chart [Source: Own Elaboration]

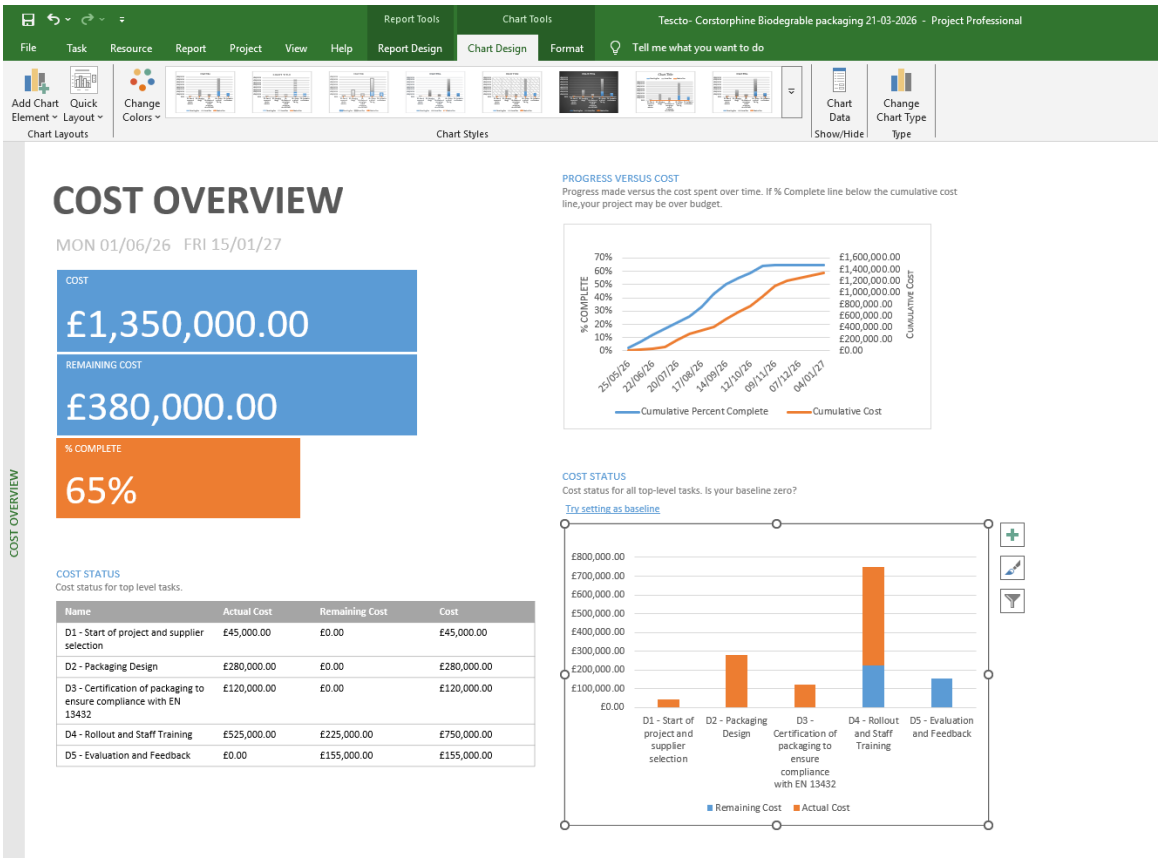


Figure 5 Projects Report: Cost Overview [Source: Own Elaboration]

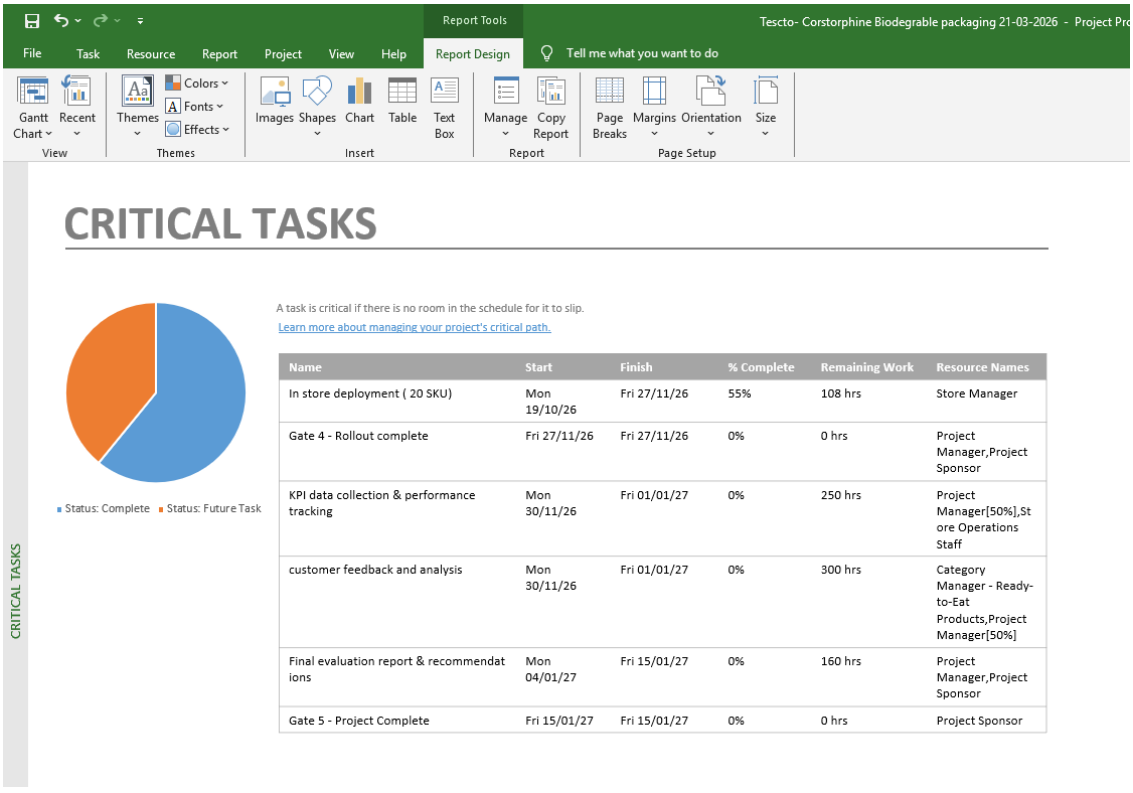


Figure 6 Projects Report: Critical Tasks [Source: Own Elaboration]

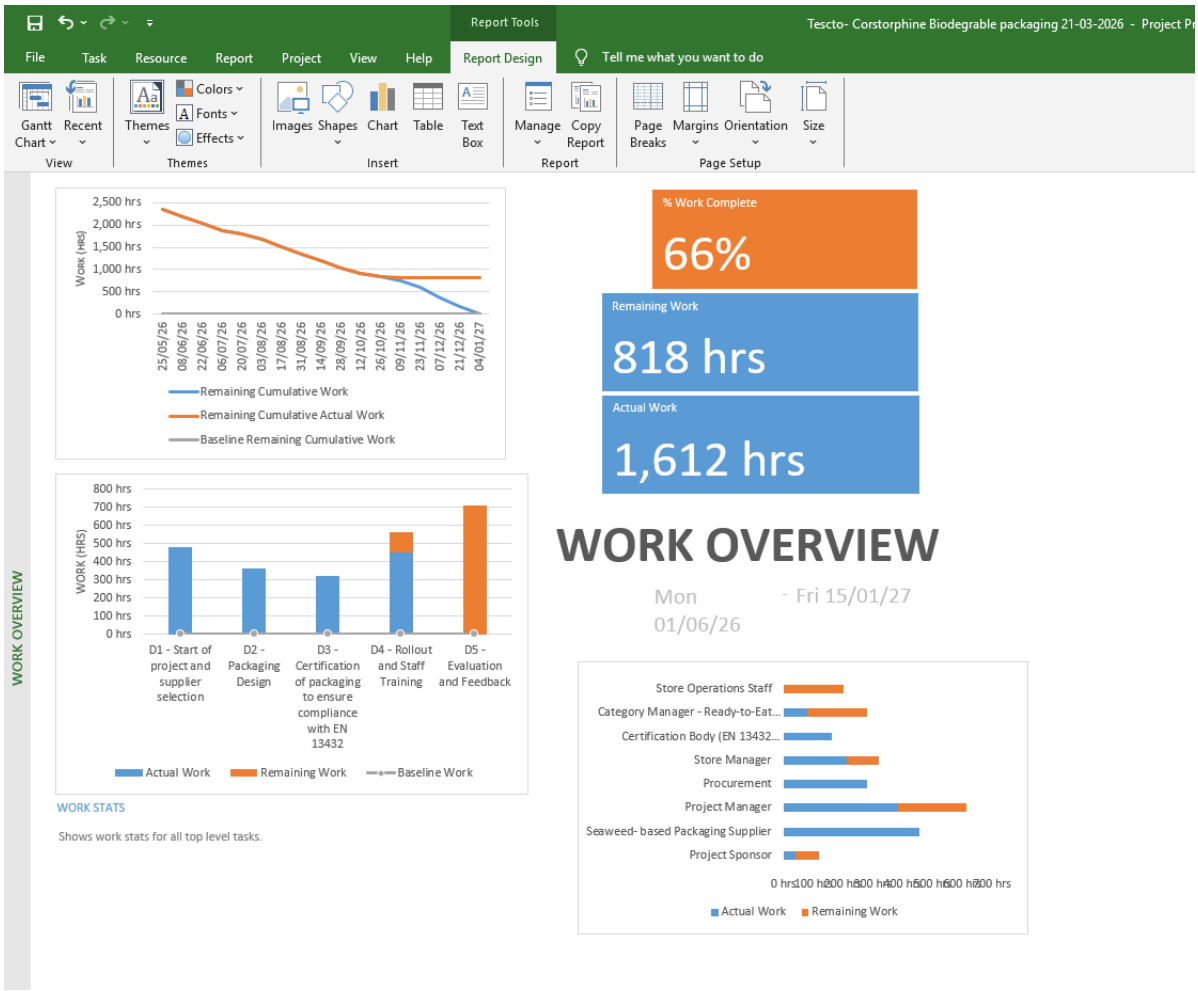


Figure 7 Projects Report: Work Overview [Source: Own Elaboration]

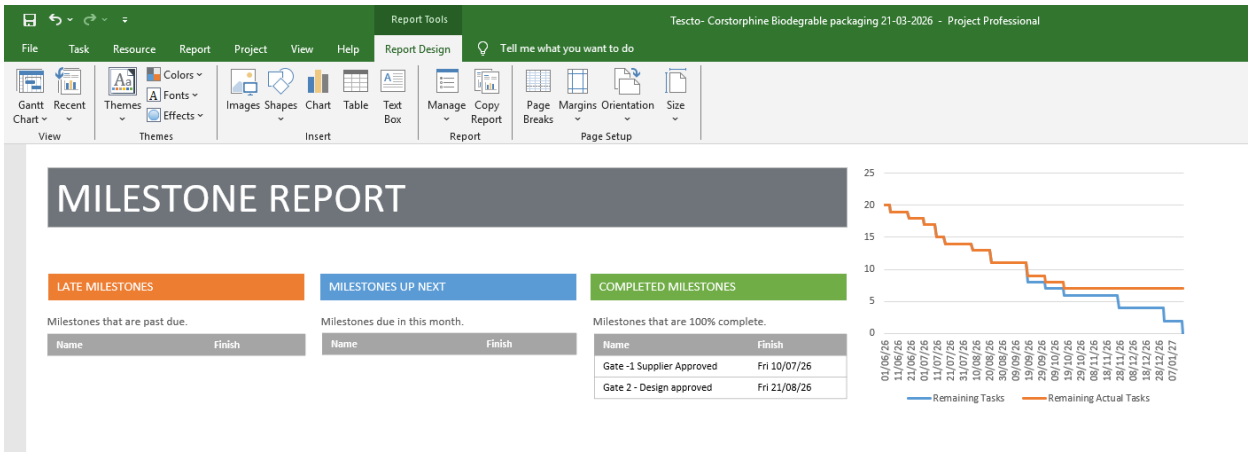


Figure 8 Projects Report: Milestone Report [Source: Own Elaboration]

Appendix 6: Stakeholder Engagement and Governance Alignment

A6.1 Responsibility Assignment Matrix (RACI)

ACTIVITY/ROLE	Project Sponsor	Project Manager	Procurement Dept.	Category Manager	Store Manager & Staff	Finance Dept.	Packaging Supplier	Waste Partner	Customers	Regulatory Authorities
Project approval & strategic alignment	A	R	I	I	I	C	I	I	I	I
Supplier selection	I	R	A	C	I	C	C	I	I	I
Packaging certification & regulatory compliance	I	R	C	C	I	I	R	I	I	A
Procurement contract & supply agreement	I	R	A	C	I	C	R	I	I	I
Store implementation & staff training	I	R	C	C	A	I	C	I	I	I
Waste handling & disposal coordination	I	R	I	I	C	I	C	A	I	C
Monitoring & performance evaluation	A	R	C	C	C	C	I	I	C	I

Table 6 RACI Matrix [Own Elaboration]

The Responsibility Assignment Matrix (RACI) is a project management tool used to clearly define roles and responsibilities for each key project activity. The RACI framework distinguishes between **Responsible (R)**, **Accountable (A)**, **Consulted (C)**, and **Informed (I)** stakeholders, ensuring that tasks are executed efficiently and decision-making authority is clearly assigned.

In this project, the RACI matrix has been applied to all major activities of the biodegradable packaging pilot, including supplier selection, packaging certification, procurement, store implementation, waste handling, and performance monitoring. Each activity has been assigned to the appropriate stakeholders, reflecting their operational and strategic involvement. For example, the Project Manager is responsible for coordinating most tasks, while the Project Sponsor holds accountability for strategic approval and performance evaluation.

A6.2 Organisational Breakdown Structure (OBS)

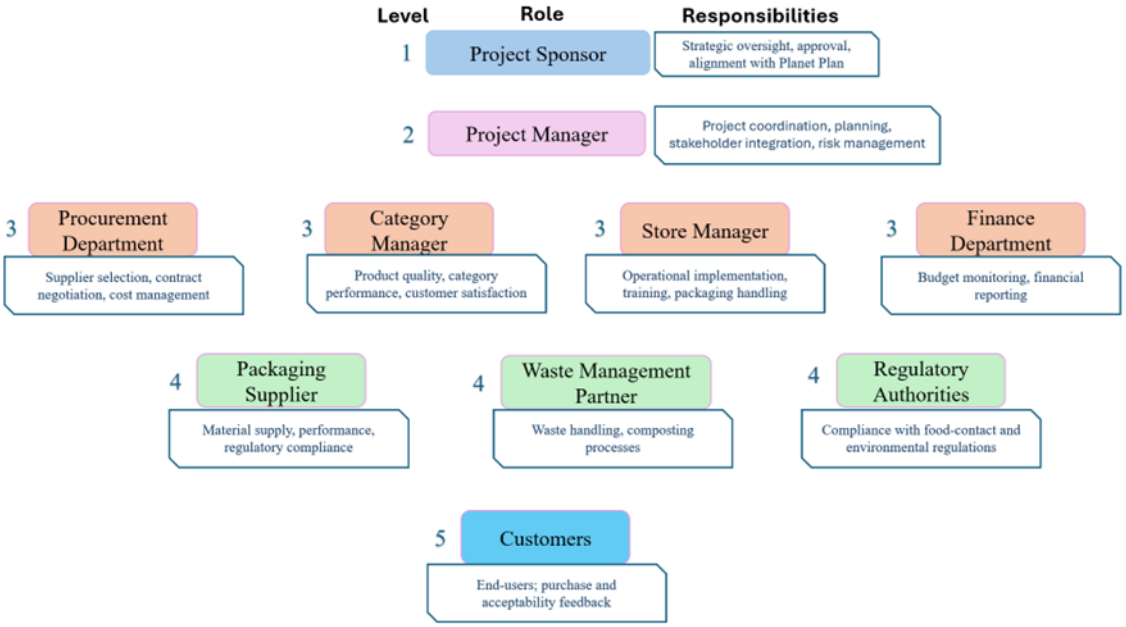


Figure 9 OBS [Source: Own Elaboration]

The Organisational Breakdown Structure (OBS) is used to visually represent the project hierarchy and clarify reporting lines among internal departments, external partners, and end users. In this project, the OBS links the project activities defined in the WBS with the responsible stakeholders, showing who is accountable and responsible for each functional area. By mapping roles and responsibilities hierarchically, the OBS supports transparent governance, efficient communication, and coordinated execution throughout the pilot implementation of the biodegradable packaging at Tesco Corstorphine Extra.

A6.3 Communication Plan

Stakeholder	Information / Message	Frequency	Channel / Method	Owner
Project Sponsor (Sustainability Director)	Project progress, budget status, KPI updates	Monthly	Steering Committee Meeting / Email report	Project Manager
Project Manager	Task assignments, progress updates, risk reports	Weekly	Project Team Meeting / Email	Project Manager
Procurement Department	Supplier status, contract updates	As needed	Email / Meetings	Project Manager
Category Manager – Ready-to-Eat	Category performance, product quality updates	Bi-weekly	Email / Briefing	Project Manager
Store Manager & Operations Staff	Implementation instructions, training guides	Before and during deployment	On-site briefing / Training session	Project Manager
Finance Department	Budget reports, cost tracking	Monthly	Email / Finance System	Project Manager
Packaging Supplier	Delivery schedules, packaging specifications	Weekly / As required	Email / Coordination Meeting	Project Manager
Waste Management Partner	Waste handling procedures, compliance info	Monthly / As needed	Email / Meeting	Project Manager
Customers	Sustainability messaging, product information	Continuous	In-store signage / Social media	Marketing / Communications

Figure 10 Communication Plan [Source: Own Elaboration]

The Communication Plan defines the who, what, when, how, and by whom of all project communications. It ensures that high-power stakeholders receive regular updates for decision-making, operational teams receive clear instructions for implementation, and external partners are informed of their responsibilities. The plan supports transparency, coordination, and engagement, linking directly with the Power–Interest Matrix, RACI, and OBS to ensure effective project governance.

Appendix 8: Detailed Risk Assessment and Scoring Matrix

A8.1 Risk Scoring Methodology

To ensure objectivity, we have applied a 5x5 Probability/Impact Matrix. The following criteria define our scoring thresholds:

1. Probability Scale (P):

- 1 (Rare): Very unlikely to happen.
- 3 (Possible): Might happen during the 8-month pilot (50% chance).
- 5 (Almost Certain): Highly likely to occur based on industry trends.

2. Impact Scale (I):

- 1 (Insignificant): Minor deviations (<£5,000) with no impact on strategic KPIs.
- 3 (Major): Clear impact on a KPI requiring management action to stay on track.
- 5 (Critical): Threat to the £1.5M budget, breach of FSA/HSE regulations, or missing the January 15th, 2027 (Gate 5) deadline.

3. Risk Rating (PxI):

- Low (1-5): Acceptable; managed through standard daily operations.
- Medium (6-10): Requires active monitoring and specific mitigation.
- High (12-25): Critical Priority (e.g., R2, R3); reviewed weekly by the Project Manager as defined in the RACI matrix.

A8.2 Risk Matrix Visualization

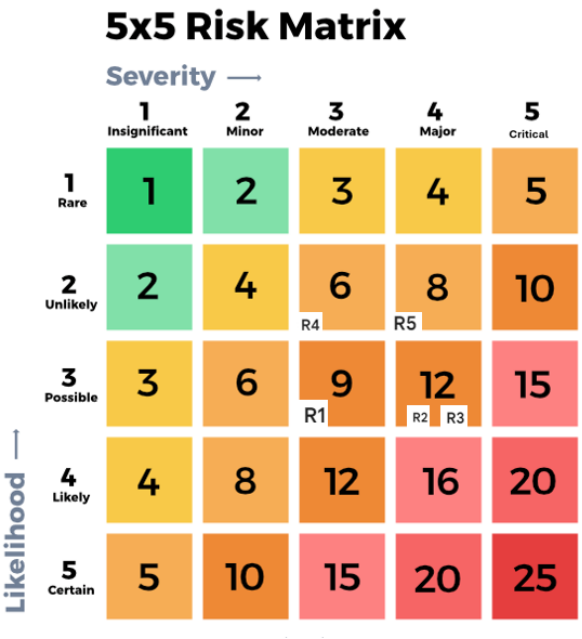


Figure 11 Risk Matrix [Source: Own Elaboration]

The matrix above visualizes the distribution of risks to facilitate resource allocation:

- Critical Zone (Red/Orange): R2 (Material) and R3 (Certification) are the top priorities with a score of 12. These are managed via the £150,000 contingency fund and schedule buffers during the critical Phase 2 and 3.
- Medium Priority (Yellow/Orange): R1 (Legal, Score 9) and R5 (Inflation/Costs, Score 8). Note that R5 is placed at Severity 4 due to its potential impact on the overall £1.5M budget.
- Low Priority (Green): R4 (Operations, Score 6) is considered low probability due to the focused nature of the pilot at Tesco Corstorphine, with mitigation focused on staff training during Phase 4.

A8.3 Detailed Risk Register

ID	Risk Event	P	I	Score	Risk Owner	Trigger (Early Warning)	Response / Contingency
R1	**Regulatory Change**	3	3	9	Tesco Legal	Official HSE notification of new SVHC restrictions.	Impact assessment & switch to pre-approved alternative seaweed strains.
R2	**Material Failure**	3	4	12	Packaging Supplier	Prototype fails moisture tests (>5% degradation) in Phase 2.	Use R&D budget for formula adjustment or move to backup supplier.
R3	**Certification Delay**	3	4	12	Project Manager	No response from EN 13432 authority by Sept 18th (1 week before Gate 3).	Escalate to Sponsor; initiate "fast-track" fee or use data from Phase 2 to overlap tasks.
R4	**Operational Issues**	2	3	6	Store Manager	>10% customer complaints or packaging breakage at Tesco Corstorphine.	In-store staff re-training and adjustment of replenishment frequency.
R5	**Supply Chain Costs**	2	4	8	Tesco Procurement	Market price of raw seaweed increases by >15% compared to baseline.	Draw from £150,000 contingency; re-negotiate bulk volume for Phase 4.

Table 7 Detailed Risk Register, Scoring, and Early Warning Triggers. [Source: Own elaboration]

This table provides full technical breakdown, scoring, and specific Triggers for all identified threats. By defining early warning signs (Triggers) for each risk, this comprehensive register ensures the project team can react instantly to deviations, protecting Tesco's £1.5M investment throughout the project lifecycle until the final completion on January 15th, 2027.

Appendix 9: Project Monitoring & Decision Dashboard.

Project Strategic Health Dashboard (Traffic Light System)

FOCUS AREA	STATUS	TREND	PM COMMENTARY
SCHEDULE	GREEN	→	On track for January 15th pilot completion. Phase 4 is 55% advanced, and data collection is ongoing.
BUDGET	AMBER	↘	Total spend is £1,200,000. Phase 3 included £70,000 contingency usage to mitigate Risk R2, leaving £80,000 for Phase 5.
QUALITY	GREEN	↑	Seaweed packaging achieving 96.2% moisture resistance, exceeding the 95% threshold for EN 13432 certification.
RISKS	AMBER	→	Major Risk R2 (Material testing failure) is Closed. Risk R3 (Supply chain capacity) is being closely monitored for the UK-wide rollout.

Figura 12 Project Strategic Health (Traffic Lights) [Source: Own Elaboration]

Context: This dashboard provides a high-level summary of the 5 main KPIs. The "Amber" status for Budget reflects the strategic decision to utilize £70,000 from the contingency fund to mitigate technical risks (R2) in Phase 3.

Deliverable	Activity / Work Package	Budgeted Cost	Actual Cost	Status
D1	Start of project and supplier selection	£ 45,000.00	£ 45,000.00	100.00%
D2	Packaging Design	£280,000.00	£ 280,000.00	100.00%
D3	Certification of packaging to ensure compliance with EN 13432	£120,000.00	£ 150,000.00*	100.00%
D4	Rollout and Staff Training	£750,000.00	£ 412,500.00	55.00%
D5	Evaluation and Feedback	£155,000.00	£ -	0.00%

Table 8 Phase Progress and Budget Tracking [Source: Own Elaboration]

Context: Detailed breakdown of task completion and financial expenditure. It confirms that Phase 4 is currently 55% complete, with the total actual cost (£1.2M) aligned with the simulated progress towards the January 2027 handover.



Figure 13 Technical KPIs and Decision Log [Source: Own Elaboration]

Context: Real-time tracking of Critical Success Factors (CSFs). The 96.2% moisture resistance rate exceeds the target, while the Decision Log provides an audit trail of formal approvals from the Sponsor and Lead Engineer.

Appendix 10:Team Contract

C11PA

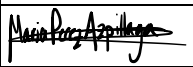
PART 2

Internal Team Member declaration form, to be submitted alongside the groupwork.

TO BE SUBMITTED IN THE APPENDIX OF THE FINAL ASSIGNMENT

To ensure that all team members have participated to the assignment, each must declare their level of contribution to the project. This is not based on number of tasks completed, but more contribution to the team and completion of the assignment. Level of contribution will be allocated to the marks. For example, if all team members have contributed fully to the project, then each member's level of contribution will be 100% and all members will receive the full awarding marks gained. However, if one team member has only contributed 50% of project and this agreed by all team members, that team member will only receive 50% of the awarded marks gained, whilst other members will receive the full 100%. See example. Each team member must complete the form electronically using Microsoft Word Application ten signed and this must be included in the appendix of the final project report. It is up to the team to discuss and agree contribution, but if no agreement can be made, the team must contact the respective tutor to arbitrate a meeting.

Team name **Group 21**

Team member	Level of Contribution (%)	Signed
Marta Armada Fernández	**100%**	
Gabriela Aragonés Fernábdez-Trapiella	**100%**	
Rocío Giménez Samaniego	**100%**	
Sambit Kuma Samanta	**100%**	
María Pérez de Azpillaga	**100%**	

Example.

Student 'A' 100%
Student 'B' 100%
Student 'C' 20%

Assuming this team is awarded 75% for the assignment, *Student 'A'* and *Student 'B'* will receive 75% towards the final mark, while *Student 'C'* would only receive 15%.

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